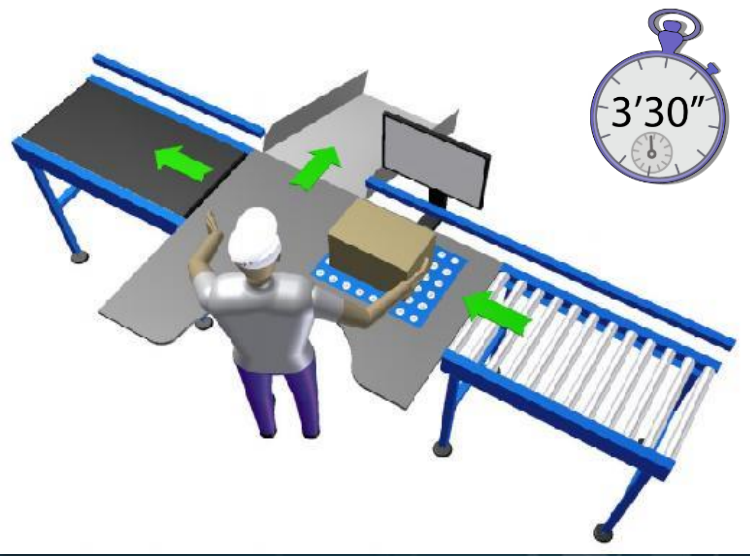




- INDUSTRIAL ORGANIZATION -

WORKSTATION STUDY Time analysis

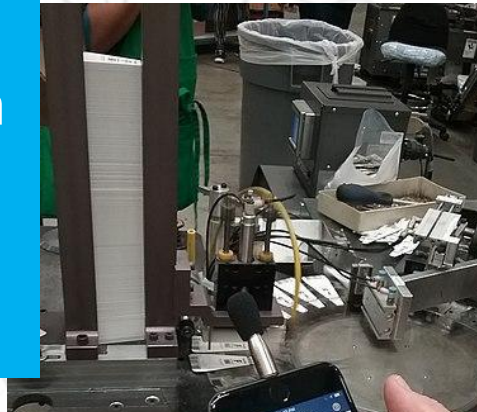
Introduction of Time and motion study



<https://youtu.be/qUgKkBhZI9c>

Background story

You are a process engineer and responsible for carrying out the following workstation study in order to suggest improvements to management



You must therefore study a car assembly workstation to improve his performance and optimize the operator's actions

Course goals

- To understand the interest and the structure of a specification
 - To know the overall method for studying a workstation
- To know the main methods of time analysis

Summary

1. **HISTORY** OF THE WORKSTATION STUDY
2. CREATION OF **SPECIFICATIONS**
3. **THE O.C.O.R.D.A.C. METHOD**, A GLOBAL APPROACH TO THE STUDY OF A WORKSTATION
4. **TIME AND MOTION** STUDY
5. **STANDARDIZED WORK COMBINATION TABLE**
6. **THE 22 RULES OF ECONOMY OF MOVEMENTS**

1. HISTORY OF WORKSTATION STUDY

1. HISTORY OF THE WORKSTATION STUDY

As early as the end of **the 19th century**, companies very quickly understood the need to define working hours for two main reasons:

- Workload assessment
- Cost of production assessment

1. HISTORY OF THE WORKSTATION STUDY

Today, the study of workstations still meets the first 2 goals:

- Valuing the **cost of production** and **valuing the workload**, but others have been added, such as:
 - Ensuring permanent progress
 - Eliminate, through organization, unnecessary gestures and non-value-added operations
 - Improve workstation ergonomics and productivity
 - Define time standards
 - Developing safety in everyday life

2. CREATION OF SPECIFICATIONS

2. CREATION OF SPECIFICATIONS

Specifications definition (source [wikipedia](#)):

The Specifications is a set of documented requirements to be satisfied by a material, design, product, process or service in a project.

There are different **types of technical or engineering specifications** (requirement specification, functional specification*, design or product specification, "in-service" or "maintained as" specification,...).

It serves to formalize the needs and explain them to the various stakeholders to ensure that everyone agrees.

Current standard: **BS 7373-1 :2001, BS 7373-2 :2001, BS 7373-3 :2005**

***NF EN 16271:** Functional expression of the need and functional performance specification

2. CREATION OF SPECIFICATIONS

APICS definition : Specification

A clear, complete, and accurate statement of the technical requirements of a material, an item, or a service, and of the procedure to determine if the requirements are met.

2. CREATION OF SPECIFICATIONS

Place of the specifications in the industrial project:

To meet the user's needs, the project engineer must realize a list of the constraints to be respected (service functions, standards, etc.), and then choose the appropriate solutions.

The success of a project depends on a written, detailed, precise, exhaustive and evaluable definition:

- the (measurable) objectives to be achieved
- the resources required
- implementation planning
- evaluation tools
- monitoring methods

2. CREATION OF SPECIFICATIONS

Place of the specifications in the industrial project:

When carrying out a project, the specifications are only one step in the next process:

- appointment of a project manager
- exploration of the technical possibilities and the state of the art
- reaching a consensus on the project within the company
- drafting of the specifications and the methodology for evaluating the bids
- issuing the specifications and formally presenting them to potential bidders
- analysis of the bids
- negotiation and signing of the implementation contract
- implementation, monitoring and evaluation of the project results

2. CREATION OF SPECIFICATIONS

2 writing approaches:

2 different points of view can be considered:

1. to impose a solution, detailed specifications, a tool or a product
2. to describe only the desired functionalities, leaving the choice of the solution to be adopted

In practice, the specifications will often be a mixture of these **two approaches**, the choice being made according to needs, the state of knowledge of potential solutions, and the level and quantity of skills available in the company to write it and analyse the offers.

2. CREATION OF SPECIFICATIONS

The call for tenders:

In the case of a call for tenders, the specifications consist of two sections:

- **a section distributed to the bidders containing:**
 - the description and context of the project
 - the applicative, technical, implementation and monitoring specifications of the project
 - administrative and contractual specifications
- **an undistributed section including:**
 - the methodology for analysing offers
 - details of the selection criteria

2. CREATION OF SPECIFICATIONS

Specification content:

1. Presentation

- a. Context (the project in brief, company description)
- b. Vocabulary/Glossary
- c. Standards to be met

2. Issues

- a. Positioning and objectives of the project (problematic, the current situation, goals, success indicator - KPI)
- b. The solution to create

3. Scope

- a. Constraints (budget, schedule, deliverables)
- b. Specific requirements (technical, implementation, administrative)

4. Organization

- a. Target audience (stakeholders, equipment, language employed, use)
- b. Project organization (date of interventions, deadline, decision-makers)

5. Deployment

- a. Training
- b. Maintenance/Support
- c. Evolution

3. **THE O.C.O.R.D.A.C. METHOD, A GLOBAL APPROACH TO THE STUDY OF A WORKSTATION**

N°	Steps		Ressources
1	Observe	1. Summary Observations 2. Consolidation and studies of the preliminary analyses	• Pareto • Product Analysis • Runtime analysis • Process Analysis • I.O. (Instantaneous Observations) • Flow Diagram • Health & safety diagnosis



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3	Observe (workstation study)	4. Detailed Observations	• Time (Chrono reading without pace time) • Standardized work combination table • I.O. • Video • Health & safety diagnosis

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3	Observe (workstation study)	4. Detailed Observations	• Time (Chrono reading without JA) • Standardized work combination table • I.O. • Video • Health & safety diagnosis
4	Reflect & Analyze (design of the new workstation)	5. Constructive criticism	• 5W - Rules • Standardized work combination table
		6. Construction or modification project	• Stabilization of labour factors • Time Estimate • MTS / MTM • Principle and practical solutions • Standardized work combination table

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5	Decide	7. 1st estimated balance sheet	• Forecast balance sheet • Profitability graph

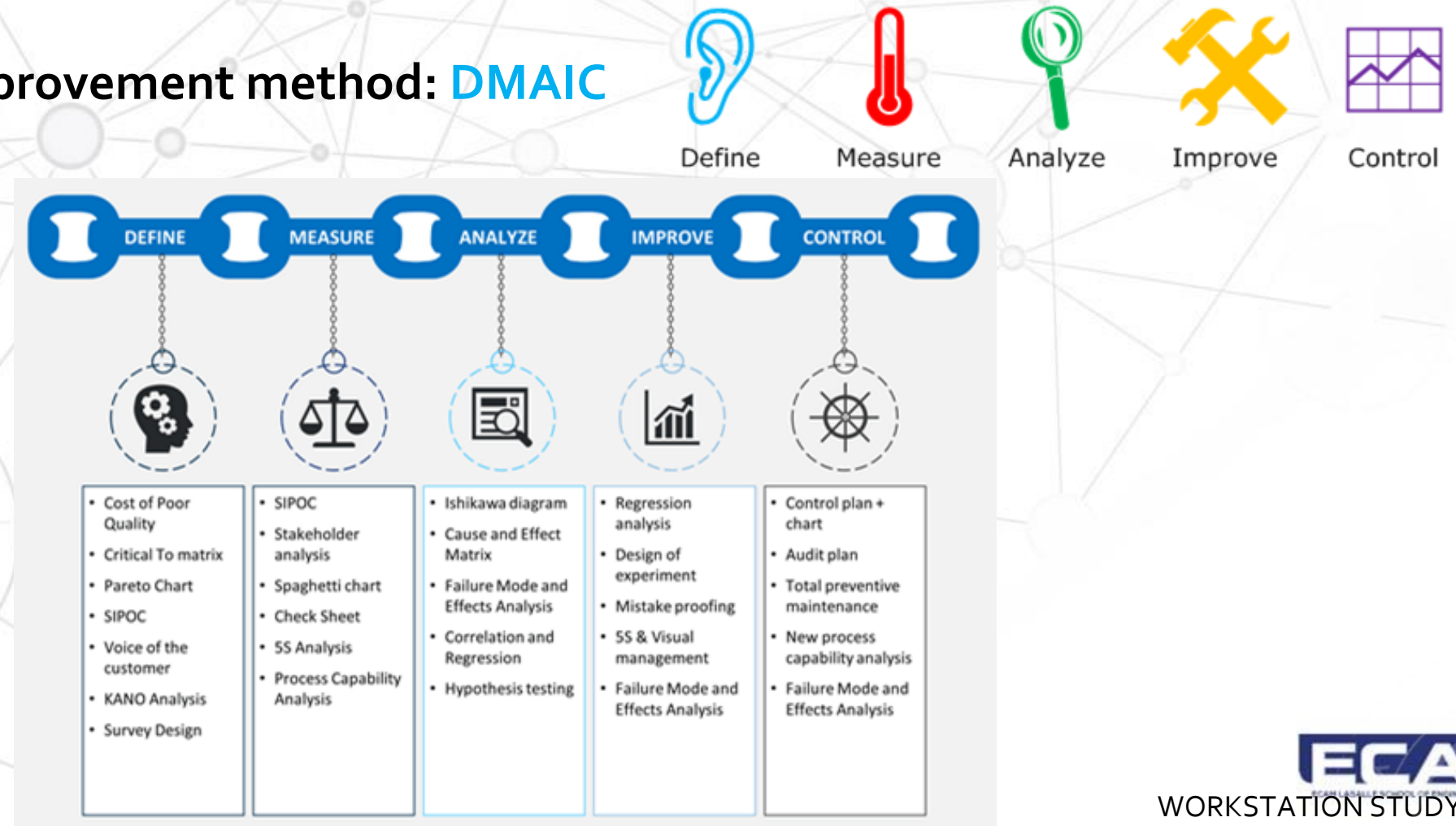
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6	Act	8. Practical implementation of the new position	• Supervision: Realization of the new post • Information of the mastery • Training of the performer • Production stabilization curve
		9. Determination of times	• Chrono readings with JA / JE • DPMA Theoretical times Th • Determination of work elements - Elementary times

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7	Control	10. 2nd final assessment	• Balance sheet • Profitability graph
		11. Control of results and settlement of disputes	• Checking the operating procedure • Timer readings with or without JA • DPMA Audit • Time confirmation or possible correction

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3. THE O.C.O.R.D.A.C. METHOD, A GLOBAL APPROACH TO THE STUDY OF A WORKSTATION

Other process improvement method: DMAIC



4. TIME AND MOTION STUDY

4. TIME AND MOTION STUDY

Time Unit:

The Time Unit systems used to the exclusion of all others are the following:

h $\Rightarrow$ time

ch $\Rightarrow$ hundredth of an hour or centihour

dmh $\Rightarrow$ ten-thousandth of an hour or decimal hour

ch $\left\{ \begin{array}{l} 1 \text{ hour} \Rightarrow 100 \text{ ch} \\ 1 \text{ mn} \Rightarrow 1,67 \text{ ch} \end{array} \right.$

dmh $\left\{ \begin{array}{l} 1 \text{ hour} \Rightarrow 10\,000 \text{ dmh} \\ 1 \text{ mn} \Rightarrow 166.66 \text{ dmh} \\ 1 \text{ second} \Rightarrow 2.77 \text{ dmh} \end{array} \right.$

	dmh	h	mn	sec
seconde	100/36	1/3600	1/60	1



4. TIME AND MOTION STUDY

The 4 main methods of determining working time:

We group these methods into two main families:

1. Direct observation techniques

- The chrono analysis
- Measurement by sampling or instantaneous observation methods

2. The pre-determination of times

- Pre-determination of times with the Methods-Time Measurement (MTM)
- Standard data

4. TIME AND MOTION STUDY

The chrono analysis

This is the simplest and most widely used process for time sheets. It is recommended to use a stopwatch graduated in ten thousandths of an hour.

The 3 forms of timing:

- Timing without judgment
- Timing with corrector
- On-the-fly timing

What do you need for a good measurement:

- The necessary equipment
- The statement sheet
- The right position

Note:

- 10 observations for cycle times less than 2 minutes is required
- A minimum of 5 observations for cycle times longer than 2 minutes is recommended

Method:

1. Select the operations and the operator to be observed
2. Cutting the work into sub-elements
3. Define the number of measures required
4. Taking measurements
5. Calculate Observed Time
6. Application of pace and efficiency factors
7. Calculation of Theoretical Time

4. TIME AND MOTION STUDY

The chrono analysis

Advantages and Disadvantages

Advantages	Disadvantages
· Allows the analyst to see the entire cycle and thus identify sources of progress · The only method to measure the actual actual time to perform a task · Quick to identify a first time base · Relatively simple to implement and explain · Quickly provides values for machine-related operations	· Requires a rate of performance based on operator skill and effort · Requires many observations on several operators · Poor performance on non-repetitive operations · Requires one person per operator to perform the analysis · Behavior or the operator can change when he is observed

4. TIME AND MOTION STUDY

Measurement by sampling or instantaneous observation methods

Quantitative analysis technique to measure in terms of time human activities, machines or any other observable activities. This method relies on a statistical study of frequencies to determine standard time. It consists of **observing work at randomly distributed moments**.

Principles:

Instantaneous and random observation, in order to deduce:

- A percentage of commitment
- With a precision fixed in advance
- With a given confidence threshold or, conversely, a certain risk of error

Means:

- Discontinuous but instantaneous observations made at random (random draw or tables of chance)
- Abaques that give the correspondence between the number of observations, the precision and the percentage of engagement
- A campaign of instantaneous observations cannot be undertaken without serious psychological preparation

Method:

1. Evaluate the desired result
2. Determine the number of observations
3. Establish and adapt the observation plan
4. Observe
5. Interpret the results

4. TIME AND MOTION STUDY

Measurement by sampling or instantaneous observation methods

Advantages and Disadvantages

Advantages	Disadvantages
· No permanent presence with the operator · Eliminates the problem of having to make constant or long-term observations · Represents the average regardless of conditions · Ideally suited for studies related to machine operations · Can be used with a performance rate	· Is based on the assumption that the operator uses an acceptable and standard method of working and without behavioural changes. · Requires analysis on many types of work · Not very profitable for only one workstation

4. TIME AND MOTION STUDY

Method Time Measurement (MTM)

Characteristics of the method:

- This is probably the most well-known predictive method of time. Widely used in industry, particularly in the automotive industry, it originated at the beginning of the 20th century.
- MTM, Method Time Measurement, is an analysis of human movements in production operations.
- Based on movement analysis tables, it is possible to predetermine operating times for each basic movement and thus deduce the total standard time of the production cycle.

Method

1. Decomposition of each of the movements of the task
2. Determination of variables for each movement
3. Coding of each movement
4. Reconstructing the times required for time events using tables
5. Determining the operation time

4.TIME AND MOTION STUDY

Method Time Measurement (MTM)

Advantages and Disadvantages

Advantages	Disadvantages
· Force to be precise in the analysis · Allows you to compare different methods of performing a new task · Encourages the simplification of movements · Eliminates the need for a performance rate · Allows to establish times in advance of production · Safe and reliable method · Allows time data to be updated easily according to method changes · Allows to measure times even on small movements · Defined a standard language for all operations	· Depends on the accuracy of the task description · Requires training time · More complex to explain to operational staff · Requires a lot of time in the analysis · Requires the chrono-analysis or standard times for "machine" times.

4. TIME AND MOTION STUDY

Standard data

The challenge of this method is therefore to **create a standard database** on all these recurring and repetitive tasks.

It thus has the advantage of:

- Provide a catalog of movement and time
- Allowing focus on improvements
- Provide a quick and reliable method for evaluating standards
- Eliminate the problem of subjective assessment of operator performance

Method

1. Decomposition of each of the movements of the task
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4. TIME AND MOTION STUDY

Standard data

Advantages and Disadvantages

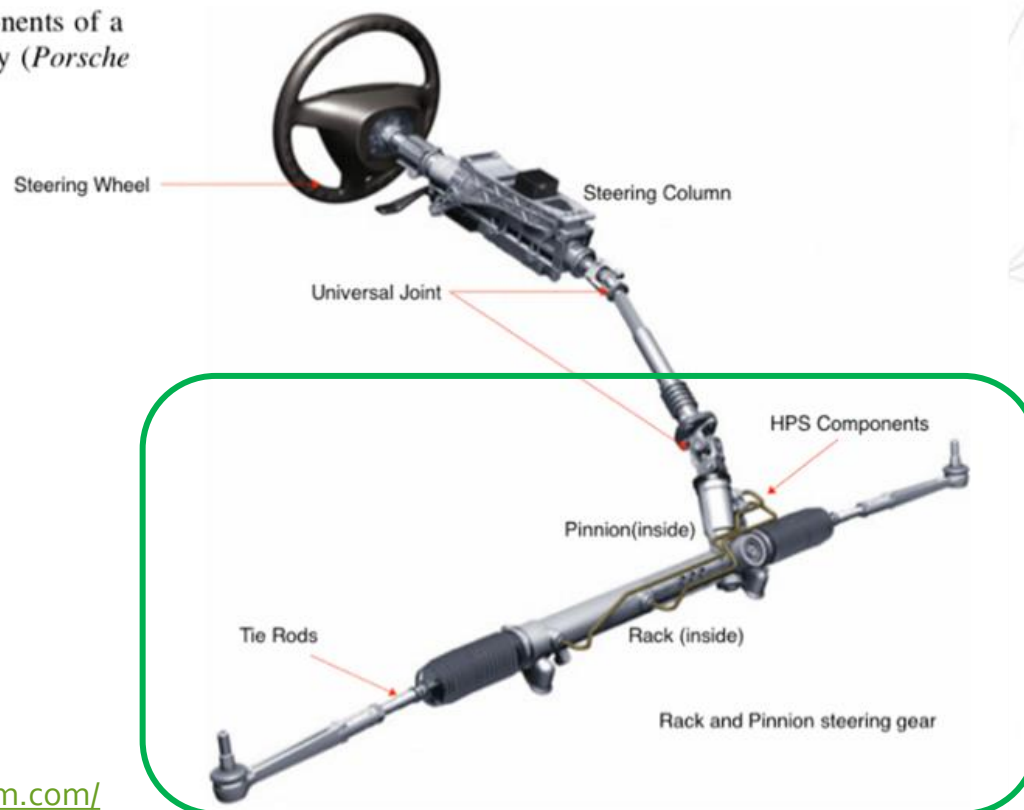
Advantages	Disadvantages
· Gives an average performance rate · Allows to establish times in advance of production · Safe and reliable method · Allows standards to be set quickly and inexpensively · Allows time data to be updated easily according to method changes	· Requires significant costs to establish the first version · Unreliable to see small variations in the change of method · Requires good analyst skills for complex formulas · Difficult to explain to operational staff

5. STANDARDIZED WORK COMBINATION TABLE

5. STANDARDIZED WORK COMBINATION TABLE

Example for the assembly of a steering system of a car

Fig. 1.1 Components of a steering assembly (*Porsche 997*)



<https://medium.com/>

Honda Prelude Mk III rear steering box
<https://en.wikipedia.org/wiki/Steering>

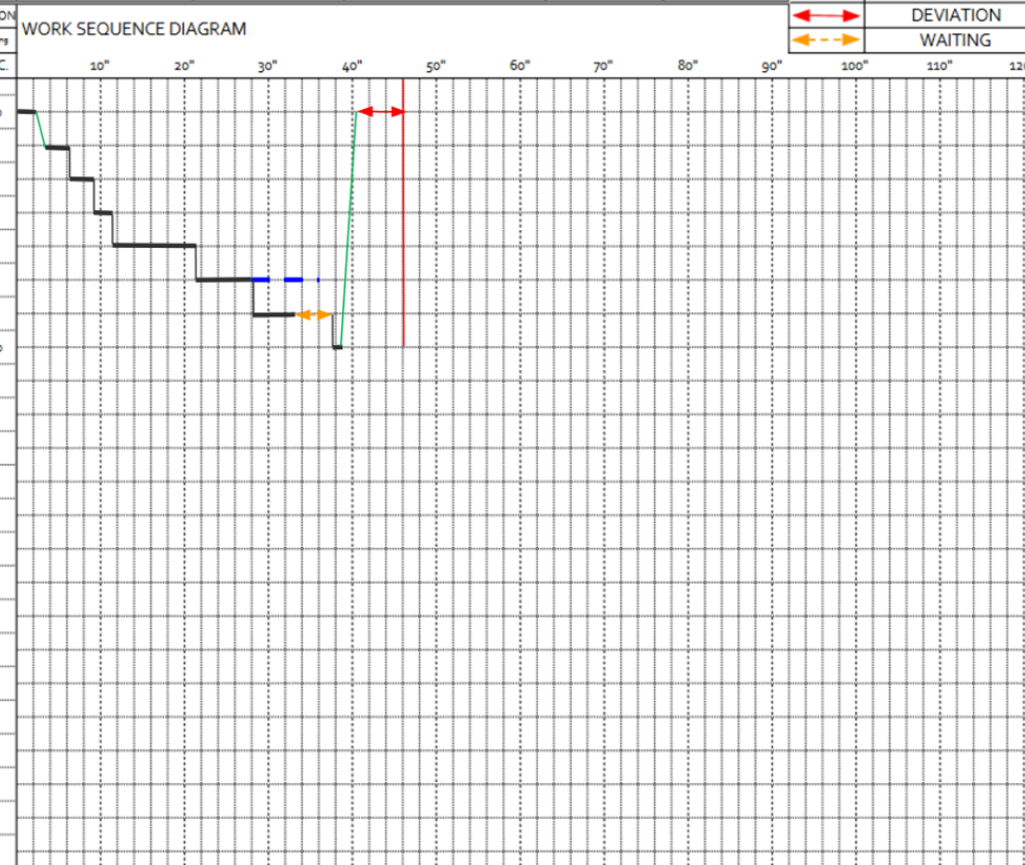


5. STANDARDIZED WORK COMBINATION TABLE

The **Standard Work Combination Table (SWCT)** combines human movement and machine movement based on takt time and is used as a tool to determine the range of work and work sequence for which a team member is responsible.

<http://www.beyondlean.com>

				OPERATOR NAME: Joe		DEPARTMENT: Method		STUDY N°: 1			
Standardized Work Combination Table				WORKSHOP: Assembly		INVESTIGATOR NAME:		Visa:			
				WORKSTATION: Steering assembly		PROCESS ANALYSIS		Takt Time: 46s		DATE: 10/08/2020	
				PROCESS PLAN:		Op. N°: 1		Cycle Time: 39,2s		———— MANUAL	
				PRODUCT DESIGNATION: Steering system		to to		Qty / shift: 100		———— AUTOMATIC	
				OPERATION NAME: Steering assembly		Op. N°: 8		Available time: 110s			
								———— WALKING			
								———— DEVIATION			
								———— WAITING			

SEQ. ORDER	WORK SEQUENCE	SEQUENCE DURATION				WORK SEQUENCE DIAGRAM
		Waiting SEC.	manual SEC.	automatic SEC.	waiting SEC.	
1	Pallet unclamping + finish part evacuation	2,4				
2	Positioning the rack on pallet N°1 + clamping	2,8				
3	Placement of the left collar	3,0				
4	Placement of the right collar	2,4				
5	Positioning of the left and right bellows	10,0				
6	Positioning of the left and right clips + Start of cycle	6,4	8,0			
7	Crimping of the left and right collars (in masked time) + Waiting for the end of the automatic cycle	5,2				
8	Back to step 1	0,8				
Cumulative Time		32,20		3,00		
Operator Cycle Time =		32,20		Nb parts / Cycle = 1,00		Cycle Time = 39,20
Deviation =		1,4		Operator waiting time = 4,00		



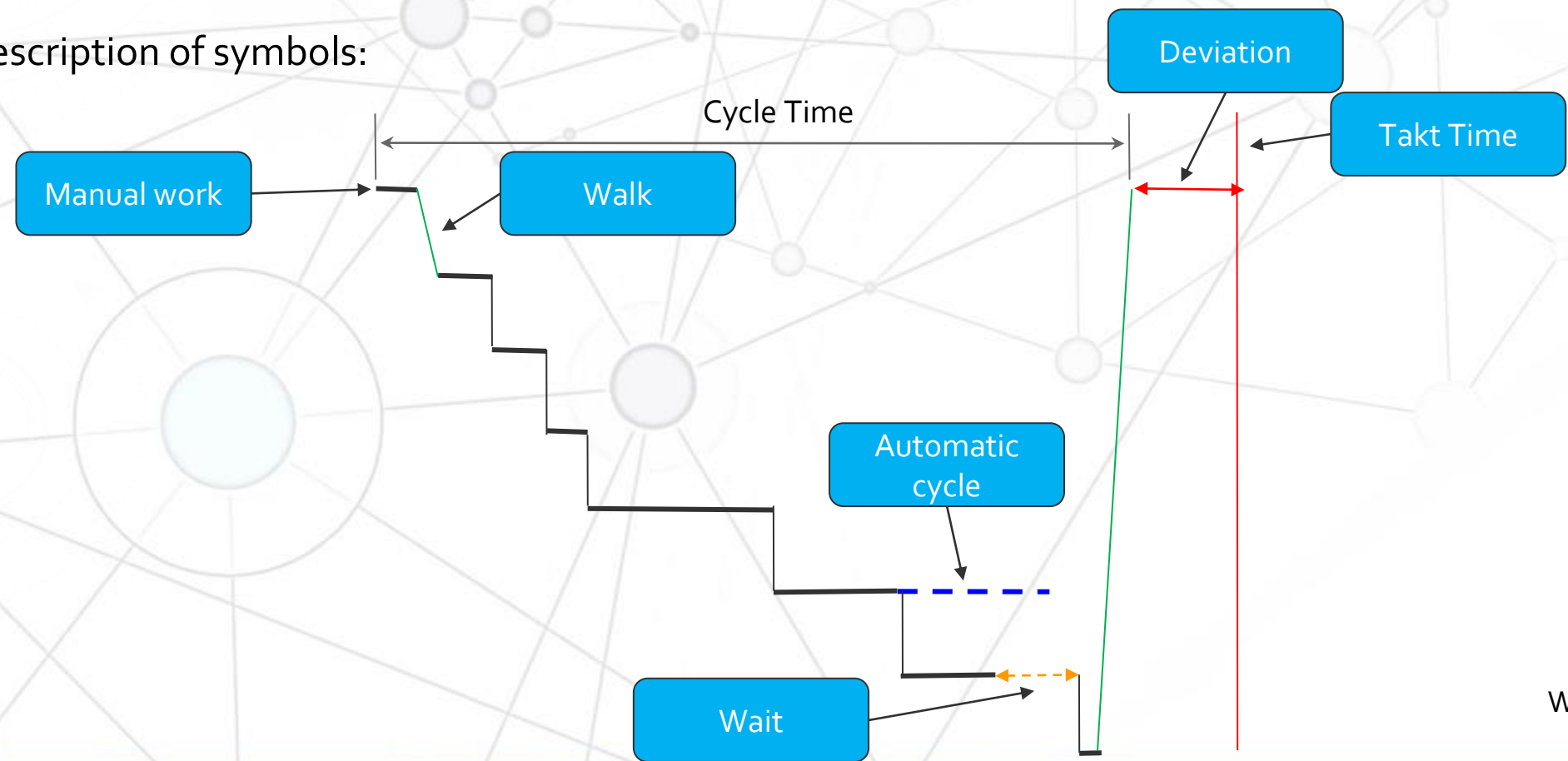
5. STANDARDIZED WORK COMBINATION TABLE

The symbols used in SWCT:

	MANUAL
	AUTOMATIC
	WALKING
	DEVIATION
	WAITING

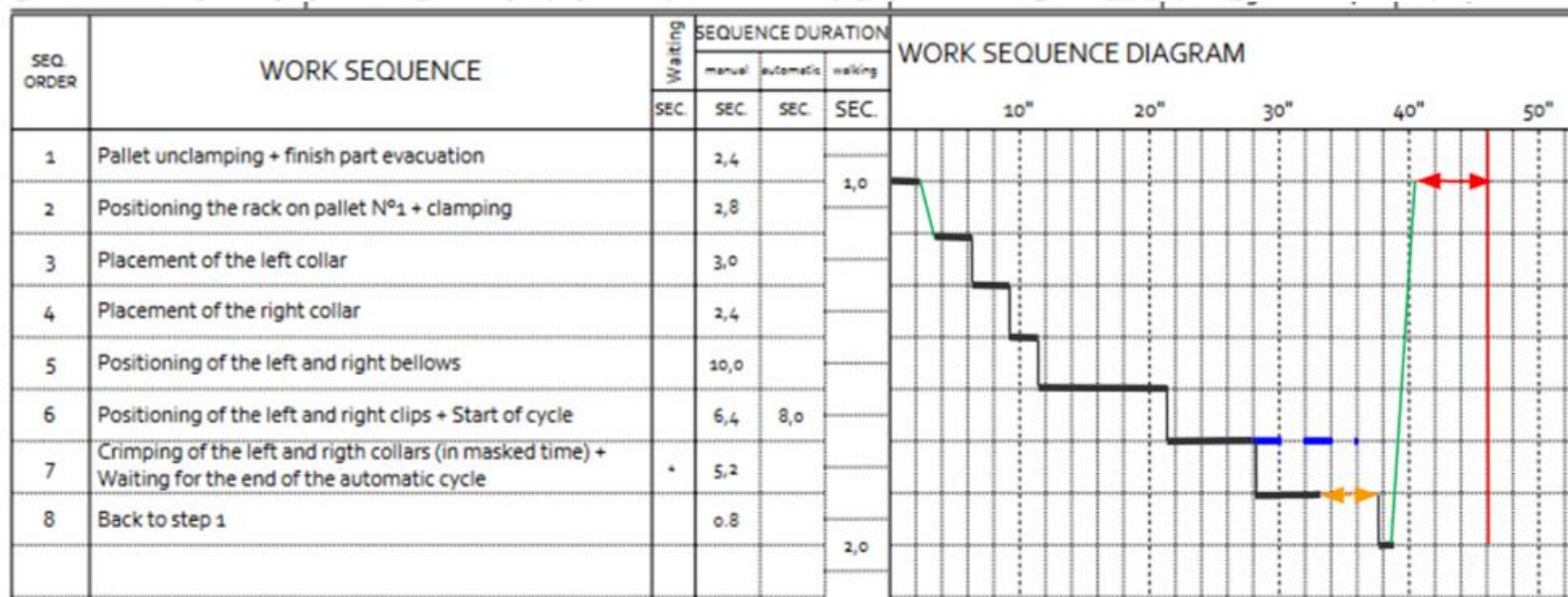
5. STANDARDIZED WORK COMBINATION TABLE

Description of symbols:



5. STANDARDIZED WORK COMBINATION TABLE

Description of work sequence:





OPERATOR NAME: Joe

DEPARTMENT: Method

STUDY N°: 1

Standardized Work Combination Table

WORKSHOP: Assembly

INVESTIGATOR NAME:

Visa:

WORKSTATION: Steering assembly

PROCESS ANALYSIS

Takt Time:

46s

DATE: 10/08/2020

PROCESS PLAN:

Op. N°: 1

Cycle Time: 39,2s

MANUAL

PRODUCT DESIGNATION: Steering system

to to

Qty / shift: 100

AUTOMATIC

OPERATION NAME: Steering assembly

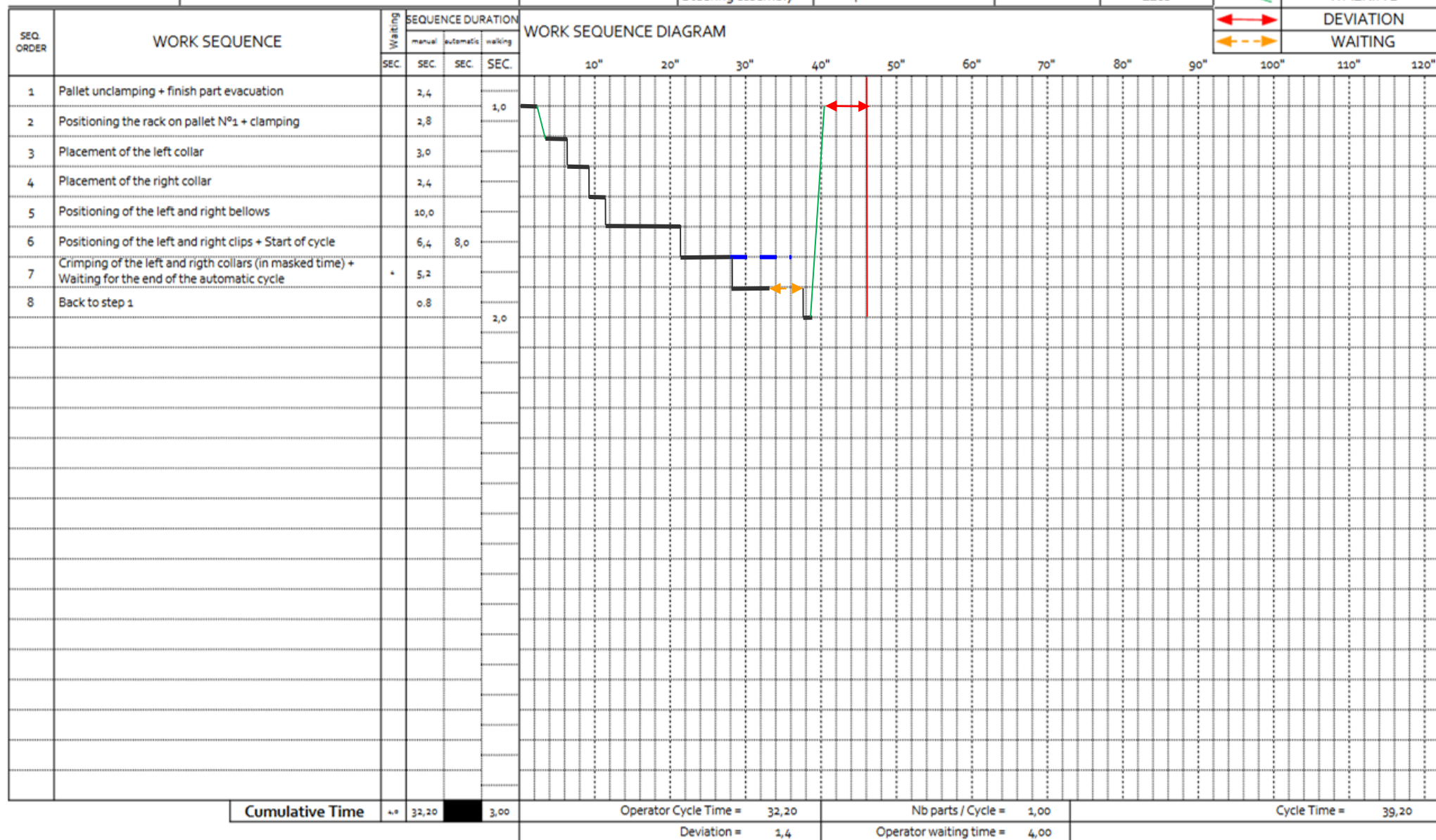
Op. N°: 8

Available time: 110s

WALKING

DEVIATION

WAITING



6. THE 22 RULES OF ECONOMY OF MOVEMENTS

6. THE 22 RULES OF ECONOMY OF MOVEMENTS

Simultaneity of movements

1. Both hands must start and end their movements at the same time
2. Both hands must not remain inactive at the same time, except during rest
3. The movements of the arms must be symmetrical and simultaneous.

Minimum energy expenditure

1. The movements necessary for work must use the smallest possible muscle mass.
2. Continuous movements are preferable to “Zig Zag” movements or movements in broken lines with acute angles

The living force

1. Live force should be used whenever possible to assist the operator's movements. It should be minimized if movement is controlled.
2. Ballistic movements are faster, easier and more precise than constrained or controlled movements.

6. THE 22 RULES OF ECONOMY OF MOVEMENTS

Pace

8. Acquiring a pace is essential to the easy and automatic execution of a Job

Order of the equipped work area

8. There must be a defined place for all materials or components.
9. Tools, materials and testers should be placed as close as possible and as in front of the operator as possible.
10. The materials, components and tools must be arranged to allow the best possible sequence of movements.

Use of gravity

8. Gravity feed boxes and containers must supply the performer(s) close to their workplace.
9. Gravity must be used for evacuation: chutes, conveyors, inclined rollers, etc.

6. THE 22 RULES OF ECONOMY OF MOVEMENTS

Comfort and lighting of the work surface

- 14. Each operator must be provided with the best conditions for the lighting of his work
- 15. The height of the work surface and the seat should as far as possible allow working standing or sitting
- 16. A seat allowing good posture must be provided to each operator

Freedom of hands

- 14. The hands must be relieved of all the work which can be done more advantageously by an assembly

Combining / positioning

- 14. Tools should be combined whenever possible
- 15. Tools and devices should be positioned whenever possible

Finger loading

- 14. When each finger performs a separate movement, the load of each finger must be distributed according to the capacities of each

6. THE 22 RULES OF ECONOMY OF MOVEMENTS

Control elements

21. The handles must allow the largest possible contact surface
22. The levers and flywheels must allow them to be maneuvered with the slightest change in posture and with as much mechanical efficiency as possible.

/!\ Quick assessment /!\

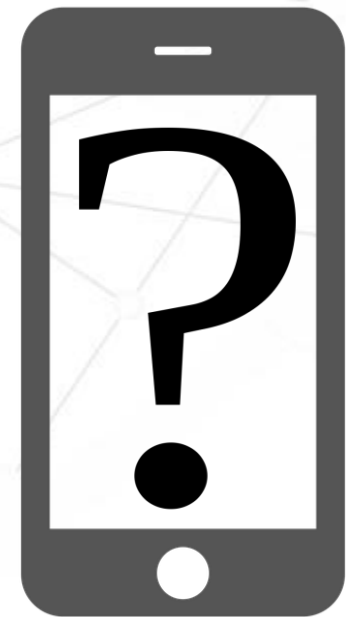
1. Go to the website with smartphone or computer:

www.menti.com

2. Add your First and LAST name (e.g. DUPONT Martin)

3. Input the code

4. Answer the questions



Thanks for your attention

DO YOU HAVE QUESTIONS?